

R/V Thompson RCA-OC Cruise TN393 - CTD Cast Log

Station: Axial Base Shallow Profile - 500m NW of Cast #/Filename: TN-393-CTD-01
 Date (Local): Aug 2, 2021 Cable Rate Out m/min: 30 m/min
 Start Time (Local/GMT): 01:11:31 End Time (Local/GMT): 2:04:25
 Start Lat/Long: 45.8333 / -129.7574 End Lat/Long: 45.8333 / -129.7574
 Surface Soak Time (mins @m): 5 min @ 10m Max Cable Out (m): 220 m
 Max Cast Depth (m): 220 m Chlorophyll Max Depth (m): ~ 50m
 Bottom Depth (m): 2649 Sampling Start/End (Local): 1919 - 2040

Niskin Bottle #	Target Depth (m)	Actual Depth (m)	Oxygen	DIC	Nutrients	Salinity	Chlorophyll	Notes
1	220		• 313A	• 101	• 3265	• 19		
2	220							
3	200			• 102	• 3266			
4	200							
5	180		• 314	• 103	• 3267	• 20		
6 ✓	180		• 315					* stopper placed upside down
7	160			• 104	• 3268			✓
8	160							
9 ✓	140		• 316	• 105	• 3269	• 21		
10	140							
11	120			• 106	• 3270			
12	120							
13	100		• 317A/31A	• 107	• 3271	• 22	✓ Cast 1 Niskin 13	
14	100							
15	80			• 108	• 3272		✓ #1 - Niskin 15	
16	80							mis fire
17	60		• 318	• 109	• 3273	• 23	• #1 Niskin 17	
18	60							
19	40			• 110	• 3274		✓ Cast 1 Niskin 19	
20	40							
21	20			• 111	• 3275	• 24	✓ Cast 1 Niskin 21	
22	20		• 319					
23	10			• 113	• 3276		✓ Cast 1 Niskin 23	
24	10		• 320					

COMMENTS:

Entered: 930 Aug 03 gn
 checked: 645 Aug 03 KCG

R/V Thompson RCA-OC Cruise TN393 - CTD Cast Log

Station:	<u>Axial Base Shallow Profiler - 500 m E</u>	Cast #/Filename:	<u>TN-393-CTD-02</u>
Date (Local):	<u>Aug 4 2021</u>	Cable Rate Out m/min:	<u>30m/min</u>
Start Time (Local/GMT):	<u>12:27:50</u>	End Time (Local/GMT):	<u>13:14:12</u>
Start Lat/Long:	<u>045.8304 / -129.7465</u>	End Lat/Long:	<u>45.8304 / -129.7465</u>
Surface Soak Time (mins @ m):	<u>5 min @ 10m</u>	Max Cable Out (m):	<u>220m</u>
Max Cast Depth (m):	<u>220m</u>	Chlorophyll Max Depth (m):	<u>~45m</u>
Bottom Depth (m):	<u>2644</u>	Sampling Start/End (Local):	<u>NONE</u>

Niskin Bottle #	Target Depth (m)	Actual Depth (m)	Oxygen	DIC	Nutrients	Salinity	Chlorophyll	Notes
1	220		.	.	.	.		
2	220							
3	200			.	.			No Samples
4	200							
5	180		.	.	.	.		
6	180		.					Bottles improperly cocked 9m
7	160			.	.			
8	160							
9	140		.	.	.	.		
10	140							
11	120			.	.			
12	120							
13	100		.	.	.	.	.	
14	100							
15	80			.	.		.	
16	80							
17	60		.	.	.	.	.	
18	60							
19	40			.	.		.	
20	40							
21	20			.	.	.	.	
22	20		..					
23	10			.	.		.	
24	10		.					

COMMENTS:

R/V Thompson RCA-OC Cruise TN393 - CTD Cast Log

Station: Axial Base Junction Box L503A

Cast #/Filename: J2-1341

Date (Local): Aug 3

Cable Rate Out m/min: _____

Start Time (Local/GMT): _____

End Time (Local/GMT): _____

Start Lat/Long: _____

End Lat/Long: _____

Surface Soak Time (mins @ m): _____

Max Cable Out (m): _____

Max Cast Depth (m): _____

Chlorophyll Max Depth (m): _____

Bottom Depth (m): _____

Sampling Start/End (Local): _____

Niskin Bottle #	Target Depth (m)	Actual Depth (m)	Oxygen	DIC	Nutrients	Salinity	Chlorophyll	Notes
1	starboard - forward		10324/325			18		
2	starboard - aft		10322/323*			*		ran out of water only 102
3								* obtained
4								suspected leak
5								when
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								
18								
19								
20								
21								
22								
23								
24								

COMMENTS:

Get collection times from entered WR Aug 6
 ✓ checked Aug 6 AM 0900

R/V Thompson RCA-OC Cruise TN393 - CTD Cast Log

Station: 500m E of Axial Base Shallow Profiler Cast #/Filename: TN-393-CTD-03
 Date (Local): Aug 4 2021 Cable Rate Out m/min: 30m/min
 Start Time (Local/GMT): 13:47:09 End Time (Local/GMT): 14:10:50
 Start Lat/Long: 45.8304 / -129.7465 End Lat/Long: 45.8304 / -129.7465
 Surface Soak Time (mins @ m): 5min @ 10m Max Cable Out (m): 220
 Max Cast Depth (m): 220m Chlorophyll Max Depth (m): ~40
 Bottom Depth (m): 2644 Sampling Start/End (Local): 0758/0850

Niskin Bottle #	Target Depth (m)	Actual Depth (m)	Oxygen	DIC	Nutrients	Salinity	Chlorophyll	Notes
1	220		.326	.125	.3277	.17		
2	220							
3	220							
4	220							
5	180		.328	.126	.3278-1	.16		
6	180		.327					
7	180							
8	180							
9	140		.329	.127	.3279	.15		
10	140							
11	100							
12	100							
13	100		.330	.128	.3280	.14	✓ Cast 3 Nisk 13	* White caps totally off
14	100							
15	60							
16	60							
17	60		.331	.129	.3281	.13	✓ Cast 3 Niskin 17	* Whitecaps to get sample
18	60							
19	40			.130	.3282		✓ Cast 3 Niskin 19	
20	40							
21	20			.131	.3283	.12	✓ Cast 3 Niskin 21	
22	20		.332/335					
23	20			.132	.3284			
24	20							

COMMENTS:

entered Aug 6 WR
 ✓ checked Aug 6 GR 0900

R/V Thompson RCA-OC Cruise TN393 - CTD Cast Log

Station: Slope Base Shallow Profiler 500m SW Cast #/Filename: TN-393-CTD-04
 Date (Local): Aug 5 Cable Rate Out m/min: 30 m/min to 200 then 60 m/min
 Start Time (Local/GMT): 23:08:29 End Time (Local/GMT): 2021-08-06 01:09:32 UTC
 Start Lat/Long: 44.5261 / -125.3945 End Lat/Long: 044.5260 / -125.3944
 Surface Soak Time (mins @ m): 5 min Max Cable Out (m): 2922
 Max Cast Depth (m): 2922 Chlorophyll Max Depth (m):
 Bottom Depth (m): 2937 Sampling Start/End (Local): N/A

Niskin Bottle #	Target Depth (m)	Actual Depth (m)	Oxygen	DIC	Nutrients	Salinity	Chlorophyll	Notes
1	<u>bottom - 10</u>							
2								<u>Bottle 2 - no water</u>
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								
18								
19								
20								
21								
22								
23								
24								

COMMENTS:

as close to bottom as possible
 * wire cut 2922 when samples taken

R/V Thompson RCA-O Cruise TN393 - CTD Cast Log

Station: Slope Base Shallow Profiler 500m SW Cast #/Filename: TN393_CTD_05
 Date (Local): Aug Cable Rate Out m/min: 30m/min
 Start Time (Local/GMT): 01:47:48 End Time (Local/GMT): 2:14:34 out of water 2:15:32 on deck
 Start Lat/Long: 044.5260 / -125.3944 End Lat/Long: 044.5260 / -125.3944
 Surface Soak Time (mins @ m): 5 min Max Cable Out (m): 220
 Max Cast Depth (m): 220 Chlorophyll Max Depth (m): 10m
 Bottom Depth (m): 2938 Sampling Start/End (Local): ~1925 - 2015

Niskin Bottle #	Target Depth (m)	Actual Depth (m)	Oxygen	DIC	Nutrients	Salinity	Chlorophyll	Notes
1	220		• 313A	• 112	• 3285			
2	200							didn't fire (issue)
3	200			• 114	• 3286	• 11		
4	200							
5	180			• 115	• 3287			
6	180							
7	160		• 314	• 116	• 3288	• 10		
8	160							
9	140			• 117	• 3289			
10	140							
11	120		• 315/319	• 118	• 3290	• 9		
12	120							
13	100		• 316	• 119	• 3291	• 8	• Cast 5	misfire
14	100		• 316	• 119	• 3291		• Cast 5	Niskin 14
15	80			• 120	• 3292	• 8	• Cast 5	Niskin 15
16	80							
17	60		• 317A	• 121	• 3293		• Cast 5	Niskin 17
18	60							
19	40		• 318	• 122	• 3294	• 7	• Cast 5	Niskin 19
20	40						• Cast 5	Niskin 20
21	20		• 320	• 123	• 3295		• Cast 5	Niskin 21
22	20							
23	10		• 321	• 124	• 3296	• 6	• Cast 5	Niskin 23
24	10			✓				

COMMENTS:

entered Aug 6 WR
 ✓ checked Aug 6 gm

R/V Thompson RCA-00 Cruise TN393 - CTD Cast Log

Station: Slope Base Junction Box L501A
 Date (Local): Aug 6 2021
 Start Time (Local/GMT): _____
 Start Lat/Long: _____
 Surface Soak Time (mins @ m): _____
 Max Cast Depth (m): _____
 Bottom Depth (m): _____

Cast #/Filename: 52-1347
 Cable Rate Out m/min: _____
 End Time (Local/GMT): _____
 End Lat/Long: _____
 Max Cable Out (m): _____
 Chlorophyll Max Depth (m): _____
 Sampling Start/End (Local): 1442-1500

Niskin Bottle #	Target Depth (m)	Actual Depth (m)	Oxygen	DIC	Nutrients	Salinity	Chlorophyll	Notes
1	2900 forward	starboard	.. 333/334			5		
2	2900 aft	starboard	.. 323/336			4		
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								
18								
19								
20								
21								
22								
23								
24								

COMMENTS:

entered Aug 28 9M ✓
 checked Aug 29 KCL 18:55 ✓

R/V Thompson RCA-O Cruise TN393 - CTD Cast Log

Station: Near dock at Slope Base DP mooring Cast #/Filename: J2-1369
 Date (Local): Aug 22 2021 Cable Rate Out m/min: _____
 Start Time (Local/GMT): _____ End Time (Local/GMT): _____
 Start Lat/Long: _____ End Lat/Long: _____
 Surface Soak Time (mins @ m): _____ Max Cable Out (m): _____
 Max Cast Depth (m): _____ Chlorophyll Max Depth (m): _____
 Bottom Depth (m): _____ Sampling Start/End (Local): _____

Niskin Bottle #	Target Depth (m)	Actual Depth (m)	Oxygen	DIC	Nutrients	Salinity	Chlorophyll	Notes
Jason 1 star	aft	2892	313			SA-0003	J2-1369	aft Niskin
2 star	forward	1500	314			SA 0002	J2-1369	forward Niskin
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								
18								
19								
20								
21								
22								
23								
24								

Star board
 4:23:59
 - aft Niskin
~~5:15:40~~
 ~ ~~2890~~ m
 2892
 - forward
 1500m
 5:15:40

COMMENTS:

entered aug 28 930 gm ✓
 checked Aug 28 18:55 KLL ✓

R/V Thompson RCA-OC Cruise TN393 - CTD Cast Log

Station: Oregon Offshore 500m E of SC013

Cast #/Filename: TNT393-CTD-06

Date (Local): Aug 26

Cable Rate Out m/min: 30 m/min 1st 200m then 60m

Start Time (Local/GMT): 9:23:24

End Time (Local/GMT): 10:20:30

Start Lat/Long: 44° 22.448 / 124° 57.013

End Lat/Long: 44° 22.448

Surface Soak Time (mins @ m): 5 min @ 10m

Max Cable Out (m): 124° 57.013

Max Cast Depth (m): 560

✓ Chlorophyll Max Depth (m): 20m

Bottom Depth (m): 571

Sampling Start/End (Local): 0330 - 0430

Niskin Bottle #	Target Depth (m)	Actual Depth (m)	Oxygen	DIC	Nutrients	Salinity	Chlorophyll	Notes
1	560					SA 0361	CTD#6 Niskin 1	
[2]	560					SA 0362		
-3	400					SA 0365	CTD#6 Niskin 3	
-4	400							
✓ [5]	200		315	DIC 136	3297	SA 0363	CTD#6 Niskin 5	
-6	200							
-[7]	180			DIC 135	3298			
-[8]	160			DIC 134	3299			
✓ [9]	140		316	DIC 133	3300	SA 0364		
-[10]	140							
-[11]	120			DIC 140	3301			
-[12]	120							
-[13]	100		317	DIC 139	3302	SA 366	CTD#6 Niskin 13	
-[14]	100							
-[15]	80			DIC 138	3303		CTD#6 Niskin 15	
-[16]	80							
-[17]	60							NO Water
-[18]	60		318	DIC 137	3304	SA 0367	CTD#6 Niskin 18	leaking noticed after DIC
-[19]	40			DIC 144	3305		CTD#6 Niskin 19	Collection
-[20]	40							
-[21]	20			DIC 143	3306	SA 0368	CTD#6 Niskin 21	
✓ [22]	20		319/320					
-[23]	10				3307		CTD#6 Niskin 23	
-[24]	10							

COMMENTS:

chlorophyll max ~ 20m

entered
checked

0930 Aug 28 gmv
19:55 Aug 29 KGL

R/V Thompson RCA-07 Cruise TN393 - CTD Cast Log

Station: Axial Base DP 1 Km E of

Date (Local): Aug 31 2021 / Sept 01

Start Time (Local/GMT): 19:54 / 04:54

Start Lat/Long: 45° 49.774 / 129° 44.769

Surface Soak Time (mins @ m): 5 min @ 10m

Max Cast Depth (m): 2600m

Bottom Depth (m): 2623m

Cast #/Filename: TN-393-CTD-007

Cable Rate Out m/min: 30m/min → 200m 60m/min → 260

End Time (Local/GMT): / 07:09 Sept 01

End Lat/Long: 45° 49.774 / 129° 44.76

Max Cable Out (m): 2600

Chlorophyll Max Depth (m): 40m

Sampling Start/End (Local): 0:23 - 0:134

Niskin Bottle #	Target Depth (m)	Actual Depth (m)	Oxygen	DIC	Nutrients	Salinity	Chlorophyll	Notes
1 ✓	2600					371	CTD7 Nisk 1	Low chlor
2 ✓	2600						CTD7 Nisk 2	" "
3 ✓	2400		321/326					
4 ✓	2250							
5 ✓	2000		322			372		
6 ✓	1500		323			377	CTD7 Nisk 6	
7 ✓	1500							
8 ✓	900		324			373		
9 ✓	900							
10 ✓	750		325					
11 ✓	750							
12 ✓	500		313			369		
13 ✓	500							didn't close
14 ✓	250					370		
15 ✓	250							
16 ✓	100		314			374		
17 ✓	100		315					Sample 1
18 ✓	75		316		3308	375	CTD7 Nisk 18	med chlor
19 ✓	60				3309		CTD7 Nisk 18	Sample 2
20 ✓	60		317		3310	379		
21 ✓	40		318		3311	378	CTD7 Nisk 21	high chlor maybe double
22 ✓	40				3312	376	CTD7 Nisk 22	high chlor
23 ✓	20		319		3313	0001		
24 ✓	10		320		3314			

COMMENTS:

✓ * 2 bottles at H, m, L chlorophyll

chlorophyll max ~ 40m

entered 09-03 9m